

SAVITRIBAI PHULE PUNE UNIVERSITY

A Mini Project Report On

NOVA.lab 2.0 — Interactive Artificial Intelligence Laboratory & Clinical Decision Support System Platform

Submitted in partial fulfillment of the requirements for the degree of

THIRD YEAR OF ENGINEERING (TE Computer Engineering)

SPPU 2024 Pattern (Term - I / Term - II)

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CERTIFICATE

This is to certify that **Vaibhav Gaikwad (24CO034)**, **Hanny Jangir (24CO040)**, **Janhavi Adagale (24CO052)**, and **Joshi Aabha (24CO053)** from Third Year Computer Engineering have successfully completed their mini project work titled “**NOVA.lab 2.0 — Interactive Artificial Intelligence Laboratory & Clinical Decision Support System Platform**” at AISSMS College of Engineering, Pune, in partial fulfillment of the bachelor’s degree in Computer Engineering under Savitribai Phule Pune University.

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MINI PROJECT APPROVAL

The Mini Project entitled:

NOVA.lab 2.0 — Interactive Artificial Intelligence Laboratory & Clinical Decision Support System Platform

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ABSTRACT

NOVA.lab 2.0 is an interactive, full-stack web-based Artificial Intelligence simulation platform and Clinical Decision Support System (CDSS) designed specifically to meet the academic, practical, and experimental requirements of the Savitribai Phule Pune University (SPPU) 2024 Pattern AI Curriculum. Traditional computer engineering AI laboratories often rely on static code scripts or terminal executions that lack real-time visualization of agent state transitions, frontier memory queue expansion, heuristic evaluation functions, and rule deduction chains. This lack of visual feedback makes complex AI paradigms abstract and difficult to master.

The **NOVA.lab 2.0** platform bridges this gap by integrating five interactive simulation modules into a unified Google Labs-inspired web portal:

1. **Reflex Agent Vacuum Environment:** Simulates the PEAS framework, state perception tuples $\langle \text{Location}, \text{Status} \rangle$, and condition-action production rules with performance metrics.
2. **Tower of Hanoi State Space Graph:** Demonstrates divide-and-conquer recursion, stack activation frames, Sierpiński gasket state graphs, and optimal move proofs ($2^n - 1$).
3. **Breadth-First Search (BFS) Grid Pathfinder:** Illustrates FIFO queue frontier expansion, cycle detection via visited sets, and unweighted shortest-path optimality.
4. **A* Search Algorithm for 8-Puzzle:** Visualizes heuristic evaluation $f(n) = g(n) + h(n)$, comparing Manhattan Distance dominance against Misplaced Tiles.
5. **Part C Mini-Project (MediMind Clinical AI Suite):** A production-grade Rule-Based Expert System supporting Forward Chaining, Goal-Directed Backward Chaining, real-time patient vitals parsing, live Knowledge Base customization, and automated Electronic Health Record (EHR) prescription report generation.

Built using React 19, Node.js/Express, MongoDB Mongoose, and REST APIs, NOVA.lab 2.0 provides students and faculty with an end-to-end sandbox, interactive quiz assessments, dual-language source code (Python & Java), and detailed academic theory manuals.

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CHAPTER 1: INTRODUCTION

1.1 Overview of the Project

Artificial Intelligence (AI) education in modern computer engineering curricula requires a balanced synthesis of theoretical rigor and hands-on simulation. Under the Savitribai Phule Pune University (SPPU) 2024 pattern for Third Year Computer Engineering, students study core AI paradigms including Intelligent Agents, State-Space Search, Uninformed Search, Informed Heuristic Search, and Rule-Based Expert Systems.

NOVA.lab 2.0 is an interactive web-based AI laboratory workbench developed to provide intuitive visual simulations for these fundamental algorithms. Rather than reading abstract formulas or running headless command-line scripts, students can interactively control

environment percepts, trigger state transitions, observe FIFO queues and priority min-heaps in real time, and test production rules.

The system features five dedicated laboratory modules:

1. **Reflex Agent Vacuum Cleaner World:** Demonstrating rational action selection under the PEAS framework.
2. **Tower of Hanoi Recursive State Search:** Visualizing 3^n state spaces and call-stack frame execution.
3. **Breadth-First Search (BFS) Maze Pathfinder:** Demonstrating uniform level-by-level wave expansion and shortest-path guarantees.
4. **A* Algorithm for 8-Puzzle:** Demonstrating heuristic admissibility and Manhattan distance calculation.
5. **Part C Mini-Project (MediMind Clinical AI Suite):** A production-grade Rule-Based Expert System for clinical diagnosis featuring Forward Chaining, Goal-Directed Backward Chaining, Vitals integration, live Knowledge Base customization, and EHR Prescription Report generation.

1.2 Need for the System

Traditional practical laboratory sessions present several challenges to engineering students:

- **Abstract State Spaces:** Algorithms like A* or BFS operate on memory structures (Open/Closed lists, Queues) that are invisible during standard code execution.
- **Lack of Interactive Feedback:** Headless console prints fail to convey how changing a single tile or wall obstacle alters heuristic evaluation values $f(n) = g(n) + h(n)$.
- **Inconsistent Practical Preparation:** Students often lack access to structured academic manuals, algorithm pseudocode, dual-language implementations (Python & Java), and viva exam quizzes in a single platform.
- **Part C Mini-Project Complexity:** Developing a realistic Rule-Based Expert System for clinical diagnosis requires complex inference engine logic, conflict resolution, and user-friendly medical report generation.

NOVA.lab 2.0 solves these issues by delivering an integrated, visual, and academically comprehensive platform accessible via any modern browser.

1.3 Purpose of the Project

The primary purpose of **NOVA.lab 2.0** is to:

- Provide an open-source, full-stack practical workbench aligned with the SPPU 2024 AI syllabus.
- Enhance conceptual clarity through step-by-step graphical visualizers for all core AI search strategies.
- Demonstrate production-grade software engineering practices by building a Clinical Decision Support System (CDSS) for the Part C Mini-Project requirements.
- Offer dual-language code samples (Python 3 & Java) along with interactive viva assessment quizzes.

1.4 Scope of the Project

The scope of **NOVA.lab 2.0** encompasses:

- Full client-side web application built with React 19 and Vite.
 - Node.js/Express backend API connected to MongoDB Atlas for persistent assignment data, theory manuals, and viva quizzes.
 - Five distinct simulation workbenches with real-time state telemetry.
 - MediMind Clinical Suite featuring Forward/Backward Chaining, Vitals processing, live Knowledge Base editing, and downloadable EHR medical reports.
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CHAPTER 2: PROBLEM STATEMENT AND OBJECTIVES

2.1 Problem Statement

"To design, develop, and deploy an integrated, web-based interactive Artificial Intelligence laboratory simulator and Clinical Decision Support System (NOVA.lab 2.0) under the SPPU 2024 Computer Engineering Pattern that enables students to visually experiment with Intelligent Agents, State Space Traversal, BFS Pathfinder, A* Heuristic Search, and Rule-Based Medical Expert Systems with automated EHR report generation."

2.2 Objectives of the Project

1. **Intelligent Agent Simulation:** Implement a Simple Reflex Agent in a 2-room Vacuum Cleaner world operating on state tuples $\langle \text{Location}, \text{Status} \rangle$ to evaluate energy vs cleanliness metrics.

- 2. **Recursive State Traversal:** Model the 3^n legal state space of Tower of Hanoi and verify the optimal move lower bound ($2^n - 1$).
- 3. **Uninformed Search Execution:** Implement Breadth-First Search (BFS) using a FIFO queue frontier and demonstrate shortest-path optimality in grid mazes.
- 4. **Informed Heuristic Search:** Implement A* search for the 8-Puzzle problem, demonstrating Manhattan Distance dominance over Misplaced Tiles.
- 5. **Clinical Expert Engine (Part C):** Develop a production-grade Clinical Expert System (MediMind AI) supporting:
 - **Forward Chaining** (Data-driven diagnosis from symptoms).
 - **Backward Chaining** (Goal-driven hypothesis validation).
 - **Real-time Patient Vitals Parsing** (Temperature, HR, SpO2 auto-triggering symptoms).
 - **Knowledge Base Extensibility** (Live addition of custom IF-THEN rules).
 - **EHR Prescription Report Generation** (ICD-10 codes, lab test recommendations, and precautions).
- 6. **Academic Enrichment:** Include formatted markdown manuals, algorithm breakdowns, time/space complexity profiles, dual Python/Java code scripts, and interactive viva quizzes.

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CHAPTER 3: SOFTWARE REQUIREMENT SPECIFICATION (SRS)

3.1 Functional Requirements

Req ID	Functional Requirement Description	Target Module
FR-01	The system shall display all 5 SPPU laboratory assignment cards on a responsive home dashboard.	Home Dashboard
FR-02	The system shall provide an interactive 2-room Vacuum Reflex Agent sandbox allowing users to drop dirt and run reflex loops.	Vacuum Sandbox
FR-03	The system shall simulate Tower of Hanoi disk moves both manually and via recursive auto-solver.	Hanoi Sandbox

Req ID	Functional Requirement Description	Target Module
FR-04	The system shall visualize BFS maze pathfinding level-by-level using an exposed FIFO queue buffer.	BFS Sandbox
FR-05	The system shall evaluate A* 8-Puzzle tile sliding using $f(n) = g(n) + h(n)$ with live Manhattan distance feedback.	8-Puzzle Sandbox
FR-06	The MediMind AI module shall parse patient temperature, heart rate, and SpO2 vitals into working memory symptoms.	MediMind Suite
FR-07	The MediMind AI module shall execute Forward Chaining inference and compute dynamic rule confidence percentages.	MediMind Suite
FR-08	The MediMind AI module shall execute Backward Chaining to prove or disprove clinical hypotheses.	MediMind Suite
FR-09	The system shall allow users to add custom production rules $\text{IF } \{S_1, S_2\} \rightarrow \text{Disease}$ live into the Knowledge Base.	KB Editor
FR-10	The system shall generate printable EHR Diagnostic Prescription reports with triage badges and ICD-10 codes.	EHR Module
FR-11	The system shall render academic theory using <code>react-markdown</code> without exposing raw syntax.	Theory Reader
FR-12	The system shall allow copying Python and Java implementation code with a single click.	Code Tab
FR-13	The system shall score interactive viva quizzes and display detailed academic explanations upon answer submission.	Quiz Tab

3.2 Non-Functional Requirements

Attribute	Specification
Usability	Minimalist Google Labs aesthetic, soft sand background (<code>#fbfbf9</code>), pill buttons, clean typography.
Performance	Instant state transition updates (< 16ms render latency using React 19).
Reliability	Includes client-side fallback assignment data so cards render smoothly even when offline.
Maintainability	Modular React component architecture separating visualizers, pages, and API hooks.
Compatibility	Fully compatible with modern web browsers (Chrome, Edge, Firefox, Safari).

3.3 Hardware Requirements

- **Processor:** Dual-Core Intel Core i3 / AMD Ryzen 3 or higher.

- **RAM:** Minimum 4 GB (8 GB recommended for concurrent React/Vite development).
- **Storage:** 500 MB free space.
- **Display:** Standard 1366x768 or Full HD 1920x1080 resolution.

3.4 Software Requirements

- **Operating System:** Windows 10/11, Linux, or macOS.
- **Runtime Environment:** Node.js v18.0+ & npm v9.0+.
- **Frontend Framework:** React 19, Vite, Lucide-React, React-Markdown, Remark-GFM.
- **Backend Framework:** Node.js, Express.js, Cors, Dotenv.
- **Database:** MongoDB Atlas / Mongoose ORM.
- **Version Control:** Git & GitHub repository.

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CHAPTER 4: SYSTEM ANALYSIS AND ARCHITECTURE DESIGN



4.1 System Modules Breakdown

1. **Navigation & Dashboard Module:** Serves the main landing page with interactive lab cards, difficulty badges, and visual previews.
 2. **Interactive Simulation Engine:** Manages local state for Vacuum, Hanoi, BFS, 8-Puzzle, and Medical Expert System.
 3. **MediMind Clinical Decision Support Engine:** Houses the patient vitals parser, forward/backward inference loops, live rule builder, and EHR generator.
 4. **Academic Theory Engine:** Uses `react-markdown` with GitHub Flavored Markdown (`remark-gfm`) to parse latex math, code blocks, tables, and blockquotes.
 5. **Assessment & Viva Module:** State-tracked quiz engine evaluating student choices against SPPU viva questions.
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CHAPTER 5: AI ALGORITHMS & CLINICAL INFERENCE ENGINE

5.1 Simple Reflex Agent & PEAS Framework

The Vacuum Cleaner world models a rational reflex agent in a 2-room environment (\$A, B\$).

- **PEAS Specification:**
 - **Performance Measure:** \$+10\$ per clean room, \$-1\$ per movement, \$-10\$ per illegal action.
 - **Environment:** 2-room grid $[A, B]$, fully observable, deterministic.
 - **Actuators:** Stepper motors $[Left, Right]$, Vacuum turbine $[Suck]$, Idle state $[NoOp]$.
 - **Sensors:** Dirt sensor $[\text{Clean}, \text{Dirty}]$, Location encoder $[A, B]$.
- **Condition-Action Rules:**
$$\begin{aligned} & \text{IF } \text{Status} == \text{Dirty} \rightarrow \text{Suck} \\ & \text{IF } \text{Location} == A \wedge \text{Status} == \text{Clean} \rightarrow \text{Right} \\ & \text{IF } \text{Location} == B \wedge \text{Status} == \text{Clean} \rightarrow \text{Left} \end{aligned}$$

5.2 Tower of Hanoi Recursion & State Graphs

- **State Vector:** $S = (p_1, p_2, \dots, p_n)$ where $p_i \in \{A, B, C\}$. Total states $= 3^n$.
- **Recurrence Relation:** $T(n) = 2T(n-1) + 1 \implies T(n) = 2^n - 1$.
- **Fractal Topology:** The state space graph forms a Sierpiński Triangle.

5.3 Breadth-First Search (BFS) Pathfinder

- **Frontier:** FIFO Queue storing trajectories.
- **Explored Set:** Visited hash set preventing cycles.
- **Optimality Proof:** Expands nodes in strict non-decreasing order of depth d . First goal hit is guaranteed minimal.
- **Complexity:** Time $O(b^d)$, Space $O(b^d)$.

5.4 A* Search for 8-Puzzle

- **Evaluation Function:** $f(n) = g(n) + h(n)$.
 - $g(n)$: Actual path cost from start to node n .
 - $h(n)$: Manhattan Distance heuristic: $h(n) = \sum_{i=1}^8 \left(|r_i - r_{\text{goal}}| + |c_i - c_{\text{goal}}| \right)$
- **Admissibility:** $h(n) \leq h^*(n)$ (never overestimates). Manhattan Distance dominates Misplaced Tiles.

5.5 MediMind Clinical Expert System (Part C Mini-Project)

- **Forward Chaining Engine:** Data-driven inference.
$$\text{Working Memory } W \subseteq \text{Symptoms} \quad \text{Rule } R_k: \text{IF } \{S_1, S_2, \dots, S_m\} \rightarrow \text{Disease } D_k$$
$$\text{Confidence} = \frac{|W \cap \{S_1, \dots, S_m\}|}{m} \times 100\%$$
- **Backward Chaining Engine:** Goal-driven hypothesis testing.
- **EHR Generator:** Produces ICD-10 coded prescriptions with recommended lab investigations and care precautions.

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CHAPTER 6: DATABASE AND KNOWLEDGE BASE SCHEMA

Mongoose Schema (`server/models/Assignment.js`)

```
import mongoose from 'mongoose';

const assignmentSchema = new mongoose.Schema({
  title: { type: String, required: true },
```

```

slug: { type: String, required: true, unique: true },
category: { type: String, required: true },
difficulty: { type: String, enum: ['Easy', 'Medium', 'Hard'], default: 'Medium' },
visualizationType: {
  type: String,
  required: true,
  enum: ['vacuum', 'hanoi', 'bfs', 'eight-puzzle', 'expert-system']
},
manual: {
  aim: { type: String, required: true },
  objectives: [{ type: String }],
  theory: { type: String, required: true },
  algorithm: [{ type: String }],
  complexity: { time: String, space: String }
},
code: {
  python: { type: String, required: true },
  java: { type: String, required: true }
},
quiz: [{
  question: { type: String, required: true },
  options: [{ type: String, required: true }],
  correctIndex: { type: Number, required: true },
  explanation: { type: String }
}]
}, { timestamps: true });

export default mongoose.model('Assignment', assignmentSchema);

```

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CHAPTER 7: API ENDPOINTS AND BACKEND IMPLEMENTATION

Express Server Endpoints (`server/index.js`)

- `GET /api/assignments` : Returns light summaries of all 5 laboratory assignments.
- `GET /api/assignments/:slug` : Returns full manual, theory markdown, Python/Java source code, and quiz data for a specific assignment.

```

app.get('/api/assignments/:slug', async (req, res) => {
  try {
    const assignment = await Assignment.findOne({ slug: req.params.slug });
    if (!assignment) return res.status(404).json({ error: 'Assignment not found' });

```

```
    res.json(assignment);
  } catch (err) {
    res.status(500).json({ error: 'Server error retrieving assignment' });
  }
});
```

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CHAPTER 8: GRAPHICAL USER INTERFACE

The interface features a **Google Labs-Inspired Light Theme**:

- **Palette:** Warm Sand background (#fbfbf9), Crisp White Cards (#ffffff), Slate Typography (#0f172a), Professional Blue Accents (#2563eb).
- **Background:** Soft ambient floating mesh gradients (.blob-1 , .blob-2 , .blob-3).
- **Controls:** Pill-shaped action buttons with Lucide icons.

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CHAPTER 9: IMPLEMENTATION DETAILS

Project Directory Structure

```
nova-lab/
├─ client/
│  ├─ index.html
│  ├─ src/
│  │  ├─ main.jsx
│  │  ├─ App.jsx
│  │  ├─ index.css
│  │  ├─ components/
│  │  │  ├─ Navbar.jsx
│  │  │  ├─ VacuumVisualizer.jsx
│  │  │  ├─ HanoiVisualizer.jsx
│  │  │  ├─ BFSVisualizer.jsx
│  │  │  ├─ EightPuzzleVisualizer.jsx
│  │  │  └─ MedicalExpertVisualizer.jsx (MediMind AI)
│  │  └─ pages/
│  │     ├─ Home.jsx
│  │     └─ AssignmentDetail.jsx
└─ server/
   └─ index.js
```

```
├─ seed.js
├─ models/
  └─ Assignment.js
```

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CHAPTER 10: CONCLUSION AND FUTURE SCOPE

10.1 Conclusion

NOVA.lab 2.0 successfully delivers a comprehensive, interactive AI laboratory platform tailored for the SPPU 2024 Computer Engineering pattern. By combining visual sandbox execution, formal theory manuals, dual-language source code, and an advanced Clinical Decision Support System (MediMind AI), the system converts complex theoretical concepts into an intuitive educational experience.

10.2 Future Scope

1. **LLM API Integration:** Connect MediMind AI directly with OpenAI GPT-4o / Gemini 1.5 Pro APIs for natural language medical consultations.
 2. **3D Visualizers:** Implement Three.js 3D renderings for 8-Puzzle and Hanoi environments.
 3. **Cloud Deployment:** Host the platform on Vercel (Frontend) and Render/AWS (Backend MongoDB).
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